A Comparative Study on Genetic Algorithm and Ant Colony Optimization in Resource Location Optimization

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***Abstract*—The resource location optimization (RLO) is a typical combinatorial optimization problem, which has a wide application background. The locations of facilities are optimized in order to minimize the users' total costs in terms of different considerations. A mathematical model for this kind of problems is established. Both genetic algorithm (GA) and ant colony optimization (ACO) are meta-heuristic evolutionary methods, and they are applicable to resolve the problem of resource location optimization. Then, which method, GA or ACO, is fundamentally more suitable to RLO? To answer this question, in this paper, a comparative study on these two methods are carried out. The conclusion is then tested in a case study. Both methods are applied for the optimization of city air terminal locations in a large city of China, which is based on the urban road network and the passenger distribution from a survey. Both methodological theory and experiences show that the ACO could achieve a better solution in solving RLO problems.**

Keywords—genetic algorithm, ant colony optimization, resource location optimization, comparison, city air terminal

I. Introduction

The resource location optimization (RLO) is a typical combinatorial optimization problem, which is widely used for solving the problems about the placement of certain facilities, so that the demands of users are satisfied and the total cost is minimized. These kinds of problems are very common in routine life. The possible facilities could be plants, depots, stores, fuel stations, metro stations, hospitals, libraries, factories, warehouses, transport hubs, etc.[1]. The facilities should be carefully localized, so that the users could access to them with minimum costs, which may include the transportation distance, the time cost, the expense, the capacity limitations, and other constraints.

The RLO methods should treat different categories of problems, including distribution of transportation system infrastructure, general services and facilities, large-scale emergency, waste management system, etc.[2] One of the first comprehensive studies about the facility location theories is given in a bibliography by Domschke and Drexl [3]. A various of problems and methods have been introduced for solving the problem of locating multiple facilities on a network to minimize an objective function with several considerations. These methods could be grouped into three main branches, which are exact, heuristic and meta-heuristic algorithms. For a complicated problem or a model of a large size, the meta-heuristic technique is widely applied in recent years, such as Tabu search, particle swarm optimization (PSO), simulated annealing (SA), and genetic algorithms (GA) [2]. In [4], a statistical analysis of the SA is applied for solving the p-median problem. The GAs have been utilized for deploying emergency response mobile units in [5]. In [6], the solutions with the GA are compared to Daskin's heuristic and integer linear programming methods. In [7], the GA is compared with LocAlloc heuristic and the Lagrangean relaxation heuristic methods for large-scale emergency problems. The meta-heuristic methods have shown their advantages compared to the heuristic methods in these applications. The ACO and multi-objective GA are utilized in [8] for solving a location-routing problem in healthcare management. A preliminary comparison of the test results with the two methods has been performed.

Both genetic algorithm (GA) and ant colony optimization (ACO) are meta-heuristic evolutionary methods. The two methods are widely applied and their effectiveness and efficiency are compared for solving the traveling salesman problem (TSP) in [9]. Stützle and Dorigo [10][11] introduced how to implement the ACO to the quadratic assignment problem and presented a direct comparison between the ACO and GA. However, a comparative analysis of these two methods in solving the resource location optimization has not been as much discussed. Therefore, this is the main objective of this work.

In this paper, a comparative study of GA and ACO in resource location optimization is performed in experiences. In particular, a mathematical model of the resource location problem is established in Section II. Then, the implementations of GA and ACO and a comparative analysis of the two methods are introduced in Section III. A case study of the RLO problem is carried out in Section IV, in order to test and verify the theoretical analysis. The paper ends with some conclusions in Section V.

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II. MATHEMATICAL MODEL

All resource distribution problems could be abstracted into a mathematical model as illustrated in Fig. 1. It is based on a network $G(V, L)$, where the set V includes all nodes and L includes all links between nodes. The network could have several different meanings relying on specific problems. For example, for planning the shopping mall locations, the nodes represent the possible locations of malls or the possible customer residence areas, the links between nodes represent the routes, and the objective is to find the mall locations with the shortest average path length from main residence areas to malls. In general, such a problem could be regarded as to find the best facility locations based on user locations, with the associating links having the lowest costs. The following definitions are used in this paper.

1) The nodes set $V = \{V_i\}_{i\in[1,N_n]}$ with a total number N_n in the network. The nodes are grouped into 3 types:

- The subset $\{U_k\}_{k\in[1,N_u]}$ includes all user locations with a total number N_u. In Fig. 1, the users are illustrated by the yellow squares. In addition, $\{w_k\}_{k\in[1,N_u]}$ includes the weights of the users. Each value of w_k indicates the quantity of users located at U_k.
- The subset $\{F_l\}_{l\in[1,N_f]}$ includes all facilities linking to users, with a total number N_f.
- The rests are the ordinary nodes.

2) The set $L = \{L_j\}_{j\in[1,N_l]}$ includes all possible links between the nodes, illustrated by the blue dotted lines in Fig. 1. The total number of links is N_l. The network is recorded as a $N_n \times N_n$ matrix M_L. Each element $M_L(m,n) > 0$, with $m, n \in [1, N_n]$, represents a connection between V_m and V_n , and $M_L(m,n)$ represents the lowest cost (positive) between two nodes. If $M_L(m,n) = 0$ means no direct link between V_m and V_n. Moreover, no self-connecting link is allowed.

3) The optimal links from users to facilities are illustrated by the red lines in Fig. 1. The corresponding cost are included in the set $\{c_k\}_{k\in[1,N_u]}$. Each element c_k is the lowest cost from U_k to its nearest facility, which is expressed by

$$c_k = \min_{l\in[1,N_f]} (C_{U_k F_l}) \tag{1}$$

where $C_{U_k F_l}$ denotes the lowest cost from the user U_k to the facility F_l.

The optimization of facility locations could be described as the following minimization problem

$$\min\ (V_{OF}),\ \text{with}\ V_{OF} = \sum_{i=1}^{N_{obj}} \alpha_i C_i \tag{2}$$

where N_{obj} is the total number of the components considered in the objective function, C_i represent the terms considered for the optimization, and α_i are the predefined weights. Thus, optimizing the facility locations is to find their locations where their links to users have the lowest cost.

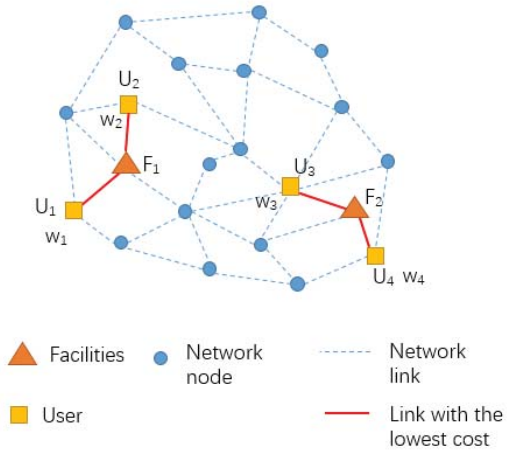

Fig. 1. Illustration of the model for the resource location optimization (RLO).

III. METHOD DESIGNS AND THEORETICAL ANALYSIS

In this section, a GA and an ACO are designed for optimizing the facility locations. A comparative study of the two methods is followed. In both GA and ACO, the ripple-spreading algorithm (RSA) in [12] is utilized as internal optimizer to find for the shortest paths between users and facility locations, as the RSA is efficient and appropriate for solving a many-to-many path optimization problem.

A. GA implementation

The genetic algorithm (GA) is an iterative procedure that maintains a population of candidate solutions. The individuals of each generation are evolved using specific genetic operations. Multiple optimal solutions could be obtained and the method is computational efficient. The chromosome structure and the genetic operations should be properly defined specific to the resource distribution problem.

1) Chromosome

The structure of one chromosome is illustrated in Fig. 2. One chromosome contains N_f genes and each gene records the node number of one facility F_l . Each generation contains N_I chromosomes. The fitness of a chromosome $ch_{p,q}$ (i.e., the p-th chromosome of the q-th generation, with $p \in [1, N_I], q \in [1, N_G]$) is defined as follows, based on the objective function value of the solution represented by this chromosome

$$\mathrm{F}_{fit}(ch_{p,q}) = \frac{1}{V_{OF}(ch_{p,q})}. \tag{3}$$

2) Genetic operations

Genetic operations are applied on the chromosomes in one generation to produce the next generation. These operations include inheritance, mutation, crossover, and natural selection.

a) Elite inheritance (application possibility p_d): The chromosomes are sorted according to their fitness values. The best $p_d N_I$ chromosomes are directly inherited.

b) Mutation (p_m): $p_m N_I$ chromosomes are randomly chosen. For each chromosome, one gene is randomly chosen and mutated.

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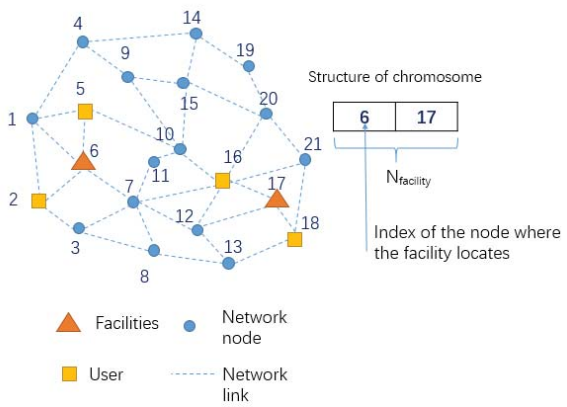

Fig. 2. Illustration of the structure of one chromosome.

c) Uniform crossover (p_c): $p_c N_I$ chromosomes are generated. One offspring chromosome is generated by two parent chromosomes which are randomly chosen by referring to fitness.

d) Random re-initialization (p_{rg}): $p_{rg} N_I$ chromosomes are randomly generated.

e) Random inheritance ($1 - p_d - p_m - p_c - p_{rg}$): The rest chromosomes are obtained by random inheritance.

By repeating the previous genetic operations, the method progressively finds individuals of higher fitness and finally the optimal or sub-optimal solutions could be achieved.

B. ACO implementation

The ACO is an agents-based metaheuristic search technique for the solution of difficult combinatorial problems, and it is inspired by the pheromone trail laying behavior of real ant colonies. The agents in the method are called artificial ants. The operations are different to those in the GA.

1) Ant's walk

Each ant chooses N_f facilities and records their node ID numbers to form a solution. The walk route of an ant is recorded by a structure the same as a chromosome introduced in Section III.A.1, which is made of discrete units. Each unit records a facility location. Each generation contains N_I ants. The fitness value of a solution is defined identically as in (3).

2) Operations of ACO

a) Pheromone-based selection (application possibility p_h): $p_h N_I$ ant-walk routes (AWRs) are generated, in which the facility locations are selected according to their pheromones laying on the network nodes. The details are given in the next subsection (III.B.3).

b) Random selection (application possibility p_{rs}): $p_{rs} N_I$ AWRs are randomly generated regardless to the pheromone information to add randomness.

c) Elite selection (application possibility $p_e = 1 - p_h - p_{rs}$): The solutions represented by the AWRs are sorted according to their fitness values. The best $p_e N_I$ solutions are directly selected into the next generation.

3) Pheromone-based selecting process

The process of the selection based on the pheromone information is described as follows.

a) Pheromone update

The pheromone of the network node V_i is increased when there are ants choosing V_i as one of their facility locations. In each generation, the pheromones of network nodes are updated.

Initially, the pheromones of all nodes are set to zero, which means that all nodes have an equal probability for being chosen as the facility locations for the initial generation. Then, as the method iterates, each ant leaves pheromone trails at the nodes chosen by this ant. If one unit of AWR (e.g. the ant of number $\mathrm{i}_a \in [1, N_I]$) contains the i_n-th node ($\mathrm{i}_n \in [1, N_n]$), the pheromone information $\tau(\mathrm{i}_n)$ of this node is updated by

$$\tau(\mathrm{i}_n) = \tau(\mathrm{i}_n) + \eta\, W_{ant}(\mathrm{i}_a)/V_{OF}(\mathrm{i}_a) \tag{4}$$

where $V_{OF}(\mathrm{i}_a)$ is the objective function value of the solution represented by the i_a-th ant. In each generation, the fitness values of all solutions associated to the ants are sorted in an ascending order. The value of $W_{ant}(\mathrm{i}_a)$ is weighted by the sequence number of the i_a-th ant's solution. Thus, $W_{ant}(\cdot)$ is an integer in $[1, N_I]$. Therefore, there are two factors that contribute to the pheromone information: one factor is the current rank of the ant in the generation, and the other is its objective function value. And η is the parameter which adjusts the separate contributions of two factors. To avoid unlimited increase of pheromone and allow forgetting of "bad" choices, a pheromone evaporation parameter ρ ($0 < \rho < 1$) is multiplied to all pheromones at each generation.

b) Possibility of selection

When one ant makes a choice for each unit of its AWR, two different ways are defined.

- Randomly choosing a location

For each unit, it has a possibility p_{rsl} for randomly choosing a facility location regardless the pheromone information. Note here, the definition of p_{rsl} is different to $\mathrm{p_{rs}}$ (defined in Section III.B.2.b). This is because $\mathrm{p_{rs}}$ refers to the possibility of randomly generating a whole AWR (or a solution) with N_f facilities locations, not only a unit.

- Selecting a location based on pheromone information

In this case, the facility location would be determined as a random variable with a probability of being selected. The possibility for choosing the i_n-th node is defined by

$$\mathrm{p}(\mathrm{i}_n) = \tau(\mathrm{i}_n)/\textstyle\sum_{\mathrm{i}_n \in [1, N_n]} \tau(\mathrm{i}_n). \tag{5}$$

C. A rough comparative analysis on GA and ACO

The GA and ACO algorithms are all metaheuristic random search methods. For solving the RLO, the effectiveness of two methods is studied in this section. Only the pheromone-based selection in ACO and the uniform crossover operation in GA are compared, since the other operations (i.e., the elite inheritance/selection or the random inheritance/selection) are the same in two methods as in Sections III.A and III.B. As metaheuristic evolutionary methods, the explicit calculations

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are impossible. Here, a comparative analysis for two typical cases are discussed to show the difference.

1) The possibility for the best facility locations to be selected into next generation

First, for the RLO problem, the form of the global optimal solution is $(V_{b_1}, V_{b_2}, \cdots, V_{b_{N_f}})$, where b_i is the node ID number of the best facility locations, $i \in [1, N_f]$, and $b_i \in [1, N_n]$.

Our discussion starts from the simplest case. We assume that at the k-th generation, one of the optimal facility locations (e.g. the node with ID number b_1, called the b_1-th node) starts to appear in one individual (chromosome or ant). The possibility for this node to be selected into the next generation is evaluated.

- In ACO: with the pheromone-based selection mechanism, this possibility is $\mathrm{p}(\mathrm{b}_1)$, as defined in (5).

- In GA: with the uniform crossover operation defined in Section III.A.2.c, the possibility is $\rho(ch_{\mathrm{b}_1})/2$, where $ch_{\mathrm{b}_\mathrm{i}}$ represents the chromosome containing the b_i-th node and the possibility for ch_{b_1} to be chosen as a parent chromosome of uniform crossover is $\rho(ch_{\mathrm{b}_1})$, which depends on the quotient of this chromosome's fitness value and the sum of fitness values of all chromosomes in this generation.

It is hard to directly compare the values of $\mathrm{p}(\mathrm{b}_1)$ and $\rho(ch_{\mathrm{b}_1})/2$ at the k-th generation, since the information of pheromones remained by the previous generations in ACO or the information about the performance of ch_{b_1} in the current generation in GA are required. Nevertheless, we see that in ACO, the possibility only relies on the node's own pheromone density, while in GA the possibility depends on the fitness density of the entire chromosome. Indeed, this chromosome contains the b_1-th node, but also contains other nodes which are not the best locations. Therefore, its performance may not be outstanding compared to other chromosomes, leading to a trivial value of $\rho(ch_{\mathrm{b}_1})$.

After some iterations, even though it is still hard to calculate explicit possibilities, an implicit discussion could be given. In ACO, the pheromone accumulates on the b_1-th node and the $\mathrm{p}(\mathrm{b}_1)$ should be increased with iterations. In GA, $\rho(ch_{\mathrm{b}_1})$ depends on the performance of all chromosomes containing the b_1-th node, after applying the crossover operations. The ratio of the chromosomes containing the b_1-th node, in general, should be increased with iterations.

A second situation is considered that two optimal facilities locations (e.g. the b_1-th and b_2-th nodes) appear in this generation, the possibility for both locations to be remained in the following generations is discussed.

- In ACO, the pheromones of two nodes are $\tau(\mathrm{i}_{b_1})$, $\tau(\mathrm{i}_{b_2})$. For the sake of simplicity, we assume that the pheromones of all other nodes are τ_0. The possibility of two nodes being selected into the next generation is

$$\frac{C_{N_n}^{N_f-2}\tau(\mathrm{i}_{b_{n1}})\tau(\mathrm{i}_{b_{n2}})\tau_0{}^{N_f-2}}{C_{N_{n-2}}^{N_f}\tau_0{}^{N_f}+C_{N_{n-1}}^{N_f-1}\left(\tau(\mathrm{i}_{b_{n1}})+\tau(\mathrm{i}_{b_{n2}})\right)\tau_0{}^{N_f-1}+C_{N_{n-1}}^{N_f-2}\tau(\mathrm{i}_{b_{n1}})\tau(\mathrm{i}_{b_{n2}})\tau_0{}^{N_f-2}} \tag{6}$$

- In GA, the possibility is about $\frac{1}{4}\rho(ch_{\mathrm{b}_1})\rho(ch_{\mathrm{b}_2})$, which depends on the fitness value densities of the two parent chromosomes, which contains the b_1-th and b_2-th nodes, respectively. The discussion could be extended to the cases with multiple optimal facilities locations. The analysis is similar.

2) The possibility for the best facility locations to be selected together by one ant (or in one chromosome)

In this part, we assume that all the best locations appear separately in the individuals of one generation, as illustrated in Fig. 3. In the figure, $N_f = 3$ and the $(V_{b_1}, V_{b_2}, V_{b_3})$ are the best facility locations. They appear in different individuals (i.e., AWRs or chromosomes). The possibility for these locations to be selected and combined in one individual is evaluated.

- In ACO: as illustrated in Fig. 4, the pheromone values $\tau(\mathrm{b}_1), \tau(\mathrm{b}_2), \cdots, \tau\left(b_{N_f}\right)$ accumulate and increase with iterations, as the solutions containing these locations have better fitness values. When the solutions start to converge, the pheromones on the nodes $i_{b_1}, i_{b_2}, \cdots, i_{b_{N_f}}$ should be much higher than other ordinary nodes, as illustrated by the green bars in Fig. 4. An ant chooses each facility location based on the pheromone information of nodes. Therefore, the possibility for the best locations to be gathered into one solution depends on the pheromone $\tau(\mathrm{b_i})$ with $i \in [1, N_f]$, which are much higher than other nodes. The probability for the appearance of the solution $(V_{b_1}, V_{b_2}, \cdots, V_{b_{N_f}})$ is relatively high.

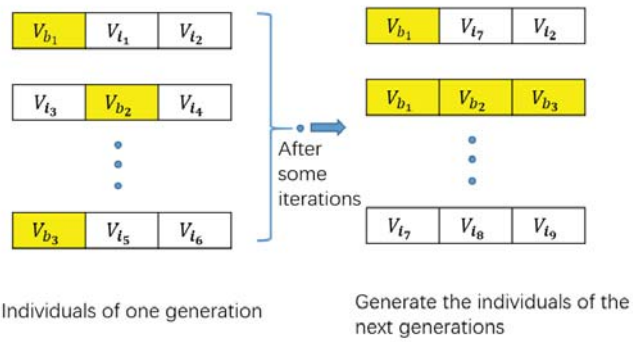

Fig. 3. All the best locations appear separately in one generation. The possibility for them to combined together in one individual is evaluated.

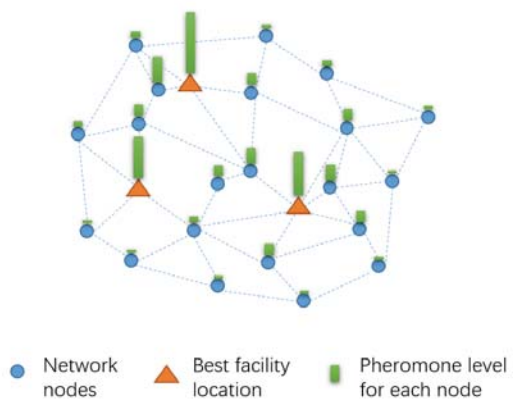

Fig. 4. Illustration of the pheromone levels after some iterations in ACO.

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- In GA: the possibility of the appearance of the solution $(V_{b_1}, V_{b_2}, \cdots, V_{b_{N_f}})$ relies heavily on the genetic operations. It seems that its capacity for combining or regrouping the good solutions is not as good as the ACO. The reason is described as follows. We could choose one chromosome with good performance as a parent chromosome, but no information is given about which gene contributes the most. All the genes are equally regarded when operating the crossover and the most "important" gene does not have a higher possibility to remain.

As a conclusion, in ACO the selection possibilities of the best locations depend on their own pheromones, which are accumulated with iterations. Therefore, they could have a greater possibility to be remained in the following generations. In GA, the mechanism is a little different, the performance of the entire chromosomes is evaluated, not the genes. In a sense, the appearance of the chromosome containing all best locations comes by "luck". We could also see that in ACO, it could be prone to falling in the local optimal solution, because it updates the pheromone according to the current best locations. But given enough iterations for the pheromones to evaporate, the ACO could find better results than the GA.

According to the previous discussion, the performance for finding the best global solution with the ACO should be better than the GA, and this will be tested in the section of case study.

IV. A CASE STUDY

In this section, a typical problem of RLO is studied, which is to optimize the locations of city air terminals in a large city. Both the GA and ACO are applied and the results are compared to show their effectiveness and efficiency.

A. Brief introduction about the RLO of city air terminals

City air terminals are small-scale terminals located in city urban areas for air transport service. Since an airport usually locates far away from downtown, the city air terminals increase the convenience by offering airline check-in, luggage-drop and airport shuttle services. To optimize the city air terminal locations, several important criteria are considered, including the average path length from passenger source locations to city air terminals, the maximum capacity volume of air terminal and the maximum passenger tolerance distance.

An illustration of this problem is given in Fig. 5. The blue dotted lines are trunk roads in the urban area. The blue circles are network nodes which correspond to the intersections or turning points of main roads. The main passenger sources are represented by the yellow squares. The objective is to locate the city air terminals, minimizing the three pre-mentioned criteria.

The case of Tianjin, China is chosen for this study. The route network of Tianjin is considered, and passenger source locations are obtained by a survey carried out at the airport. As illustrated in Fig. 6, the route network based on a real map is established. The green links correspond to the trunk roads commonly chosen by buses and taxis. Among the network nodes, the blue circles are the passenger sources and the numbers nearby are the weights of the passenger sources. The rest are the ordinary nodes where few people take an airplane

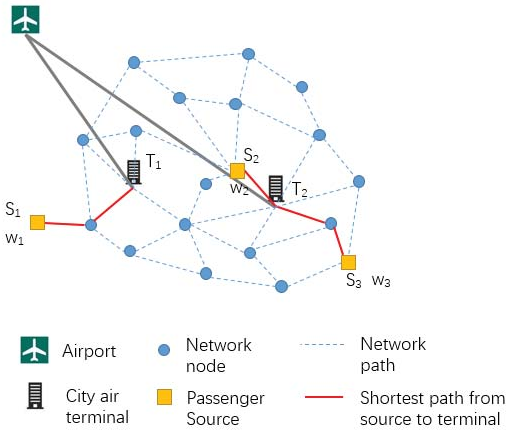

Fig. 5. Illustration of the location optimization of city air terminals in a city.

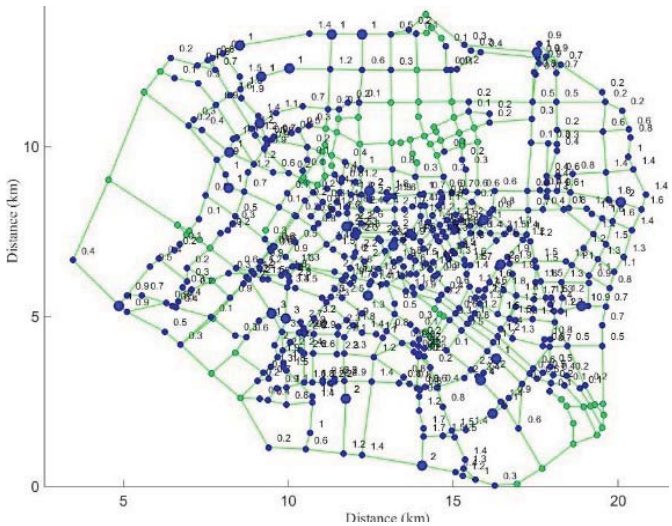

Fig. 6. Urban network with passenger sources of Tianjin, China.

from these positions. The following comparisons between the two methods are based on this case problem, which is a typical problem of RLO.

B. Tests with the genetic algorithm

The settings of model parameters include the total number of the city air terminals $N_f = 3$, the maximum tolerable distance $D_{max} = 3$km, and the passenger volume of a city air terminal $V_{max} = 201$ (the sum of the weights of all passenger sources $W_{tot} = 602.5$, so, V_{max}=W_{tot}/N_{term}).

The optimization is to find the city air terminal locations, minimizing the objective function which includes the average path length from passengers to terminals, the exceeded distance, and the exceeded volume. For the sake of simple demonstration, the weights of the three terms α_i (i = [1,3]) in the objective function are all set to 1. These parameters are the same for all experiences and could be further modified according to the requirements of the airport.

In this part, the GA with different genetic operation possibilities are tested. These different possibilities have major impacts on the optimization results and convergence speed,

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especially the possibility of mutation p_m and the possibility of uniform crossover p_c. In the tests, the range of p_m is from 0.1 to 0.3, spacing 0.05, and the range of p_c is from 0.5 to 0.7, spacing 0.05, $p_d = 5\%$ and p_{rg}=10%. The rest corresponds to the random inheritance operation. For each combination of p_m and p_c, $N_{test} = 50$ tests are performed to avoid randomness. The number of chromosomes in a generation is $N_I = 200$ and the total number of generations is $N_G = 50$.

In Fig. 7, as an example, the optimization results of the case with $p_m = 0.2$ and $p_c = 0.6$ are plotted. The top left subfigure gives the evolutions of the objective function values in 50 tests (plotted by the green lines) and the evolution of the average values (plotted by the red line). The average objective function value at the final iteration is 43.27 and the average computation time is 1131s. In the rest three subfigures, the evolutions of the three components in the objective function are plotted. At the end of optimization, the average path length from passengers to city air terminals is 3.21 km, the sum of exceeded distance compared to D_{max} is 33.39 km, and the sum of exceeded volume of all terminals compared to V_{max} is 6.67.

The results of all test cases are plotted in Fig. 8. The objective function values and three components values are listed. In all cases, the average path lengths are quite similar. However, the results of the exceeded distance and the exceeded volume are different. The case with $p_m = 0.2$ and $p_c = 0.6$ gives the best results. The results will be compared with the ACO in the next subsection.

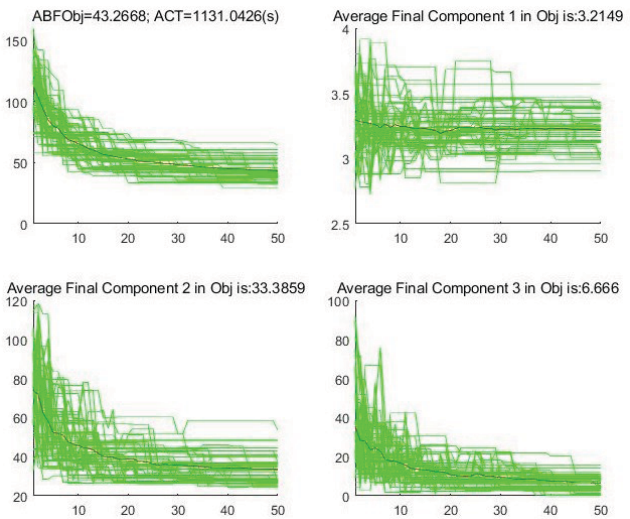

Fig. 7. Optimization results of the case with $p_m = 0.2$ and $p_c = 0.6$ of GA.

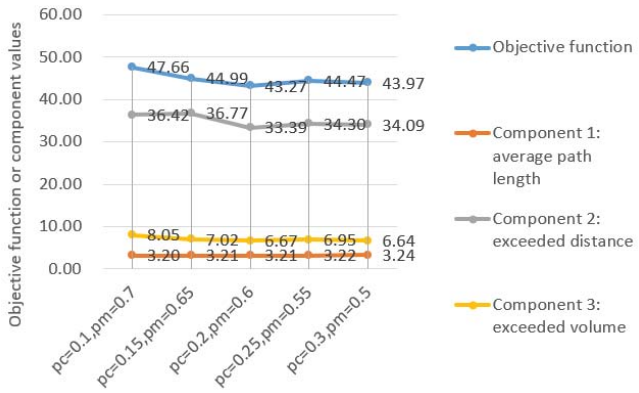

Fig. 8. Optimization results of different cases with GA.

C. Tests with the ant colony optimization

The ACO method introduced in Section III.B is applied to optimize the city air terminal locations. The parameters of ACO include $N_I = 200$, $N_G = 50$, the possibility for randomly selecting a solution $p_{rs} = 0.1$, the possibility for randomly choosing a unit of solution $p_{rsl} = 0.1$, and the possibility of elite selection $p_e = 0.05$. The parameter η which adjusts the separate contributions of two factors and the pheromone evaporation parameter ρ have influences on the optimization result and they are chosen differently in the tests. The range of η is from 0.3 to 0.7, spacing 0.2, and the range of ρ is from 0.4 to 0.8, spacing 0.2. The parameters of nine tests are listed in the first column of Table I.

The results including the objective function values, the values of the three components, and the computation times are recorded in Table I and the associating curves are plotted in Fig. 9. In all cases with the ACO, the optimization results are better than the ones with the GA in Section IV.B. The results of the first component, i.e., the average path length, are quite similar. The other two components, the exceeded distance and the exceeded volume are better than the GA. The computational efficiencies of ACO and GA are similar. The GA is slightly faster than the ACO. The best result is given by the case with $\eta = 0.3, \rho = 0.8$. The objective function value is 38.33, which is 13% better than the best solution with GA. The average path length is 3.17, which is similar to the results given by GA. The exceeded distance is 30.51, which is about 10% better than GA. The exceeded volume is 4.65, much better than GA. The evolutions of the objective functions and its three components in 50 tests are plotted in Fig. 10. From the curves of the 50 tests, we see that the solutions with the ACO converge earlier than the GA. Finally, the best optimization results of the city air terminals in 50 tests is plotted in Fig. 11. The yellow circles represent the optimal facility locations.

The results show that both the ACO and GA methods could treat well the RLO problems. The ACO method achieves a better result compared to the GA. The efficiency of the two methods are similar. The tests verified the theoretical analysis introduced in Section III.C.

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V. Conclusion

In this paper, a comparative study of the genetic algorithm and the ant colony optimization in resource location optimization has been performed. Firstly, a mathematical model of the problem has been introduced. Then, the implementations of a genetic algorithm and an ant colony optimization method have been introduced. A comparative analysis has been made. Then, a test case has been set up for locating the city air terminals based on the city center of Tianjin, China. The two evolutionary methods have been tested and compared to show the effectiveness and efficiency. The ant colony optimization has shown its advantage in solving resource location problems in both theoretical analysis and tests.

TABLE I. Test parameters and results of the ACO method

Parameters	V_{BOF}^{final}	C_1	C_2	C_3	Time (s)
Case-1 ($\eta = 0.3, \rho = 0.8$)	38.33	3.17	30.51	4.65	1262
Case-2 ($\eta = 0.5, \rho = 0.8$)	38.52	3.16	30.38	4.97	1161
Case-3 ($\eta = 0.7, \rho = 0.8$)	38.47	3.19	30.03	5.26	1179
Case-4 ($\eta = 0.3, \rho = 0.6$)	40.11	3.18	31.27	5.66	1202
Case-5 ($\eta = 0.5, \rho = 0.6$)	39.22	3.16	31.53	4.53	1187
Case-6 ($\eta = 0.7, \rho = 0.6$)	39.04	3.19	31.13	4.73	1204
Case-7 ($\eta = 0.3, \rho = 0.4$)	39.85	3.18	31.59	5.09	1312
Case-8 ($\eta = 0.5, \rho = 0.4$)	39.01	3.16	30.74	5.10	1221
Case-9 ($\eta = 0.7, \rho = 0.4$)	39.63	3.18	31.07	5.38	1203

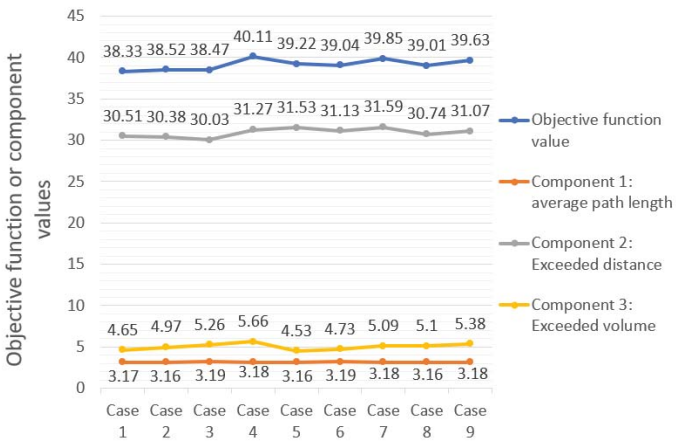

Fig. 9. Optimization results of different cases of ACO.

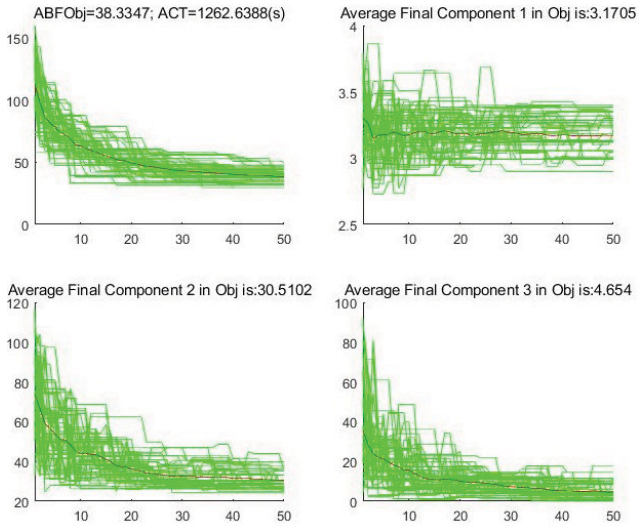

Fig. 10. Optimization results of the case with $\eta = 0.3$ and ρ=0.8 of ACO.

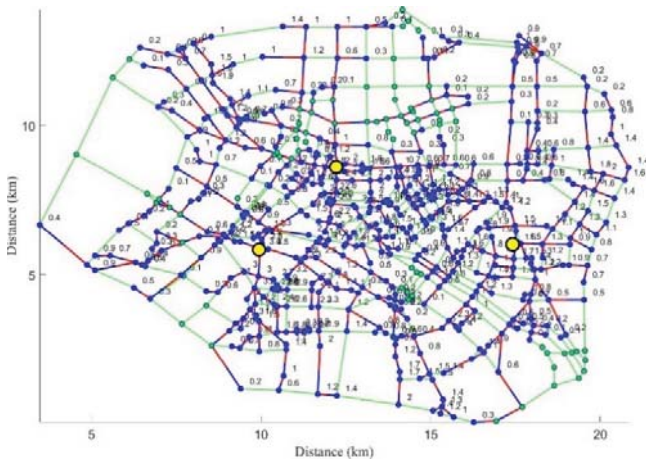

Fig. 11. Optimization result of the case with $\eta = 0.3$ and $\rho = 0.8$ of ACO.

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